



The Future is Web

By Zeeshan Ahmad Khan

About the Founder

Name : Zeeshan Ahmad Khan

Age: 31

Description: A creative and innovative individual who is very curious and passionate about science and technology, a drawing artist in childhood turn out to be a science freak in college and now a tech expert in both mechanical and computer science industry.

A big Tony Stark (Iron man idealizer)

With **3 years** of experience in Core manufacturing industry and **7 years** is software development specialized in 3D visualization on web.

Having a majority of experience working in Startups.





The Web revolution

The attention span of user gone from minutes to seconds

Increasing preference for web apps over native apps due to ease of access and updates.

Sharing content by just a link (no downloading entire app for one content)

Rise of cloud computing and SaaS models enhancing web application usage.

Example of groundbreaking trends driving this revolution : Progressive Web Apps (PWAs), online collaboration tools (e.g., Google Docs), and web games running with WebGL.

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The Evolution of Web Technologies

From Static Pages to Dynamic Web Apps

Introduction of HTML5, CSS3, and JavaScript

Game changing frameworks like React , Angular and Vue.

Rise of WebGL and WebRTC



What was missing?

Performance unlike native Apps!

Web application use Javascript which runs in browser with its compilation engine. (eg. chrome v8)

As its an Interpreted language, it is slower than compiled languages like C++ or Rust

Webgl, which is use for rendering 3D Graphics on browser is limited in performance, its built on top of OpenGL ES 2.0, a subset of the OpenGL API used in mobile devices.

Many old and effective libraries are written in C++



The game changers



WEBASSEMBLY



WebGPU



Web Assembly (WASM)

WebAssembly (WASM) is a binary instruction format for a stack-based virtual machine, designed for high performance and portability. It enables near-native speed for web applications by compiling languages like C, C++, and Rust to run efficiently in web browsers, enhancing performance, security, and cross-platform compatibility.

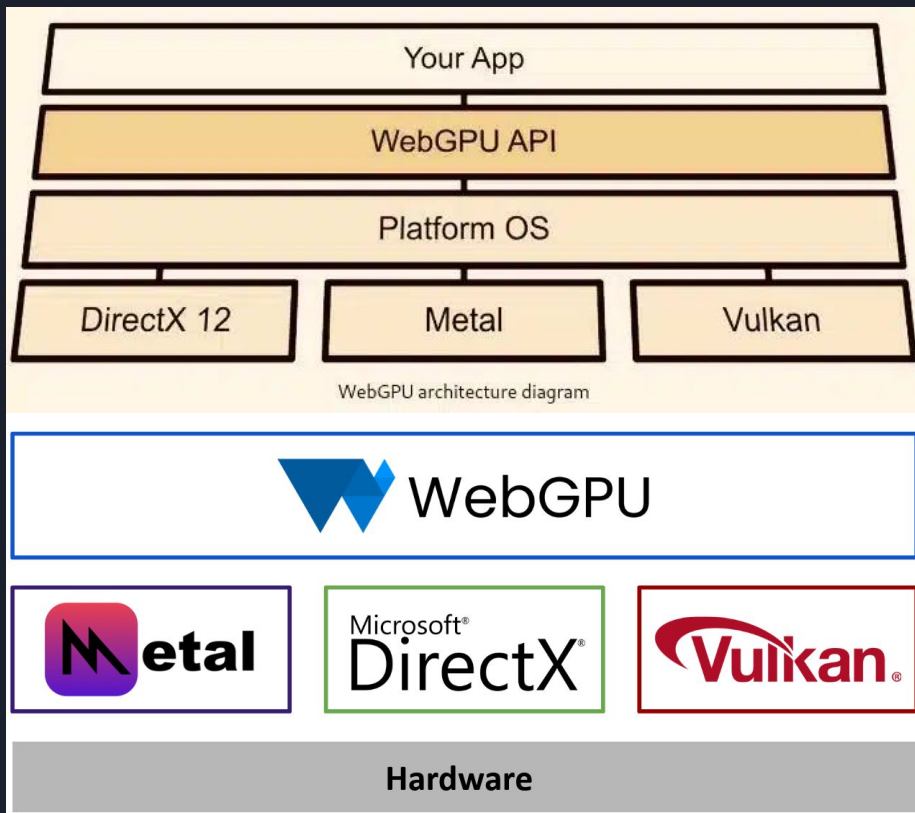
In other words you can run your C++ native code and libraries in Web browser!

Popular apps using them

Photoshop, Figma, Sketchup etc

WebGPU

WebGPU is a next-generation graphics API for web applications, offering high-performance 3D graphics and computation. It provides lower-level GPU access than WebGL, improving efficiency and performance. With support for advanced features like compute shaders, WebGPU is designed for complex graphics tasks, enabling richer, more powerful web



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So now what?

Pretty much every Software can be moved to web browser

Now application will not be downloaded but will be bookmarked on your favorite browsers .

You can do Coding, CAD Modeling, Video edition, Gaming, Video calling, Graphic designing etc all in Web browser. And share it to your friends by just a link.



Building 3D application on Web

To build a 3D Web App you can either implement webGL or WebGPU directly with javascript

Or

Use popular Libraries like **Three.js** and **Babylon.js**.

If you are using Reactjs the **react-three-fiber** is the best Library to get started.

These libraries gives you direct access to some of the web browser APIs in simplified way. Eg
Simply add VRButton from three.js Library to open your 3D scene in Virtual reality (With supported device)



Three.js

Three.js is a JavaScript library that makes it easy to create and display 3D computer graphics in a web browser. It simplifies the process of creating 3D graphics by abstracting the complexities of WebGL, the underlying technology for rendering 3D graphics in the browser.

Threejs gives easy access to WebGL, WebGPU and WebXR along with some WASM libraries of Physics and file compression engines

Simple example

```
<script>
// Scene setup
const scene = new THREE.Scene();
const camera = new THREE.PerspectiveCamera(75, window.innerWidth / window.innerHeight, 0.1, 1000);
const renderer = new THREE.WebGLRenderer();
renderer.setSize(window.innerWidth, window.innerHeight);
document.body.appendChild(renderer.domElement);

// Cube setup
const geometry = new THREE.BoxGeometry();
const material = new THREE.MeshBasicMaterial({ color: 0x00ff00 });
const cube = new THREE.Mesh(geometry, material);
scene.add(cube);

// Camera position
camera.position.z = 5;

// Animation loop
function animate() {
  requestAnimationFrame(animate);

  // Rotate the cube
  cube.rotation.x += 0.01;
  cube.rotation.y += 0.01;

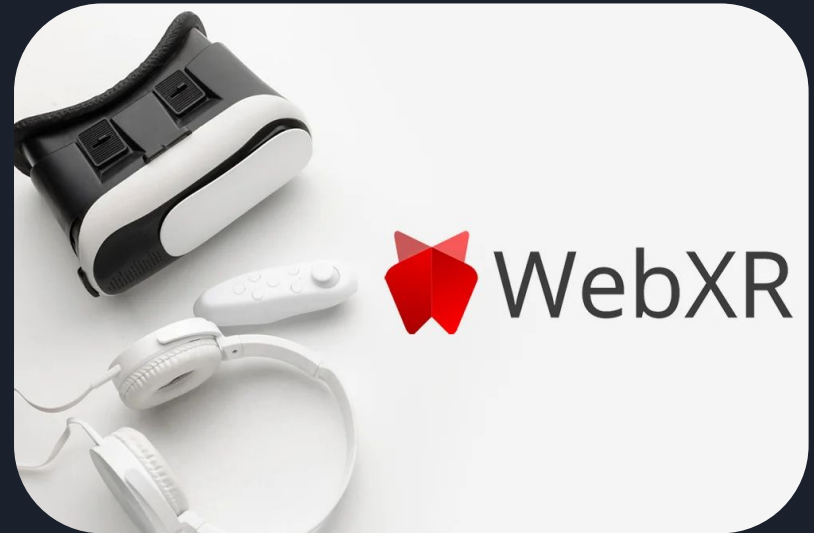
  renderer.render(scene, camera);
}

animate();
</script>
```

WebXR

WebXR, or Web Extended Reality, is an emerging standard that allows for immersive experiences to be delivered through web browsers. It expands upon the capabilities of WebVR (Virtual Reality) and WebAR (Augmented Reality) by encompassing both virtual and augmented reality experiences, as well as mixed reality.

WebXR aims to democratize XR experiences by lowering the barriers to entry. Users can access WebXR content directly through their browsers, making it more convenient and less intimidating for casual users.

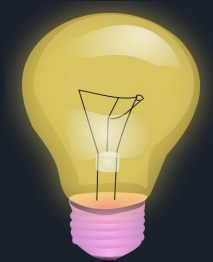


Basics of 3D development

Whether you are creating a Game, rendering video or virtual reality experience, the basic principle is the same.

3D Scene which consist of 3D meshes/objects, Lights and Camera and

properties like Materials, Textures, Animations, Sounds, Colliders and Background environment map.



Lights



3D Objects



Camera



The new possibilities with web

Just like CANVA is revolutionizing the Graphics designing industry



Our new platform Zimension3.com is simplifying the development of 3D Games and VR content Immersive Experience by providing web based Game engine.





Thank you